

GCP Intro

Saturday, April 5, 2025 12:38 PM

-----Concept of Zones, Regions and multi-region-----

Regions:

Regions are independent geographic area that consist of zones. They effect pricing, reliability, networking and performance.

42 Regions as in April 2025

Regional Resource: App Engine

Zones:

A zone is a deployment area for google cloud resources within a region. Zones should be considered a single failure domain within a region. To deploy fault tolerant applications with high availability and help protect against unexpected failures, deploy your application across multiple zones in a region.

127 Zones as in April 2025

Multi -Regional Services are designed to be able to function following the loss of a single region.

Multi Regional Resources: Big query, Big table, Cloud Storage, Spanner, Data Store, fire Store, Artifact Registry.

3 Multi Regions.

Note:

If a single region fails only customers solely in that region are impacted customers who have multiple regions products are not impacted.

Fully Qualified name for a zone is made up of <region><Zone>

For EG: Zone A in region US-Central1 is US-Central1-A

When Selecting Zones yourself, have are something to keep in mind

1. Communication within and across regions will incur different cost.
2. Design important Systems with redundancy across multiple Zones or Regions.

At some point in time your instances might experience an unexpected failure To mitigate the effects of these possible events you duplicate/replicate important resources in multiple

zones or regions.

-----Resources Hierarchy in GCP-----

The Purpose of the google cloud resource Hierarchy is twofold:-

1. Provide a hierarchy of ownership which binds the lifecycle of a resources to its immediate parent in the hierarchy.
2. Provide attaches and inheritance for access control and organization policies

GCP resources hierarchy resembles the file system found in traditional OS as a way of organizing & managing entities hierarchically. Each resource has exactly one parent.

Organization---->folders--->dept1, 2, 3----->projects----->1, 2, 3----->
Resources

Benefits of the Organizational Resource

Organization:-

With an Org Resource, projects belongs to your org instead of the employee who created the project. This means that the project are no longer deleted when an employee leaves the company.

--> You can grant roles of the org level, which are inherited by all projects and folders under the org resource.

For E.G.: you can grant the network admin role to your networking team at the org level, allowing them to manage all the networks in all company, Instead of granting them the role for you all individual projects.

Folders:-

-->Folder resource provides an additional grouping mechanism and isolation boundaries between projects.

-->Folder can contains other folders, its optional.

Project:-

-->All resources you we are associated to one specific project.

--> You can track resources & quota usage.

-->Enable billing & set budget.

-->Manage permissions and credentials.

-->Project is a global entity.

-->Enable services & API.

-->Equivalent "Account" to AWS and "Subscription" in MS Azure.

Project has three components:-

Project ID, Project Name & Project Number.

Project ID--> GLOBALLY UNIQUE, CHOSEN BY YOU, IMMUTABLE

Project Name-->UNIQUENESS NOT REQD, CHOSEN BY YOU, MUTABLE

Project Number-->GLOBALLY UNIQUE, ASSIGNED BY GCP, IMMUTABLE

Google Cloud Compute engine/Virtual Machine

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Compute Engine is a customizable compute Services that lets you create and run Virtual Machines on Google's Infrastructure.

There are two types of Machine in GCO:-

1. Custom Machine-You can take VCPU and Memroy as per your need, and create your own custom Machine.
2. Predefined Machine:-Predefined Machine types are pre-built and ready to go configuration of VM's with specific amounts of VCPU and Memory to start running app quickly

Google Compute Engine is comes under IAAS.

A VM is lime a physical Hardware server, but running in the cloud.

GCP isolates resources and sells them on a per server basis but different combinations.

Google provides VM of all sixes to micro to 160 VCPU and 3.37 TB Memory.

You can attach local SSD up to 3TB and persistent disk up to size of 64TB, that too encrypted as well.

Google Cloud uses open source KVM hypervisor at backend.

Fast Booting time among all cloud providers.

All VCPU, GCP and GB of memory are charged a minimum of 1 Minute After 1 Minute instances are changed in 1 sec increments.

--->Local SSD- DAS-Direct attached Storage, MAS-Machine attached storage. Temp disk

Types of pre-defined VM GCE

General Purpose

Compute Optimized

Memory optimized

Accelerated Optimized or GPU

Shared core

Sole Tenant Nodes

A machine type is a set of virtualized hardware resources available to a VM including the system memory size, VCP and persistent disk limits.

In compute engine, machine types are grouped and put inside families on their behavior & workload.

Compute Engine has 3 terms:-

Machine Family-Set processors and hardware configurations.

Series-Machine families are further classified by series and generations.

Machine Types-Every machine series has pre-defined machine types that provide a set of resource

Customer using GCE-Paypal twitter, P&G, Airbus ETSY, SKY.

Lec-07 - Google Cloud Compute Engine or Virtual Machines / GCE					
Workload Type					
General Purpose Workloads			Optimized Workloads		
Cost-Optimized	Balanced	Scale-Out Optimized	Memory Optimized	Compute Optimized	Accelerator Optimized
E2	N1, N2, N2D	Tau, T2D	M1, M2	C2, C2D	A2
Day to Day Computing at a Lower Cost	Balanced price/performance across a wide range of VM Shapes.	Best Performance & Cost for Scale Out Workloads	Ultra high memory Workloads offers Highest Memory Config across VM Families	Ultra high performance for Compute intensive Workload	Optimized for high performance Computing Workload Best for Parallel Computing Workload
→ Web Serving → App Serving → Back office apps → Small & Medium Databases → Microservices → Virtual Desktops	→ Web Serving → App Serving → Back office apps → Medium-large Databases → Cache → Media/Streaming	→ Scale-Out Workloads → Web Serving → Containerized Microservices → Media Transcoding → Large Scale Java Application	→ Medium-large in-memory databases such as SAP HANA → In-memory Databases and In-memory analytics → Microsoft SQL Server and Similar Databases	→ Compute Bound Workloads → High performance web Serving → Gaming → Ad Serving → High performance Computing → Media Transcoding	→ CUDA enabled ML Training and Inference → HPC → Massive Parallelized Computations

Shared Core machines

This type of VM's intends to timeshare the physical core shared core machine types are a cost effective option that works well with small or batch workloads that only needs to run for a short time. It uses partial VCPU that runs on one-hyper thread of the host CPU running your instance.

Sole-Tenant Nodes

Sole Tenant nodes are the servers for compute engine that are dedicated for the users on priority. The purpose of these nodes is to sever the deployment for BYOL(Bring your own license) Application.

The sole tenant nodes provides you access to similar VM configuration and Machines types just like regular compute instance

Custom machine type

If none of the predefined machine types matches your needs, you can independently specify the number of CPU's and the amount of memory for your instance.

The memory per VCPU of a custom machine types must be between 0.9 GB, 3.75 GB per VCPU.

Total memory for a custom Machine type must be a mulitple of 256 MB

Above 1 the instance VCPU count must be even such as 2 4 6 8 and so on

Compute Engine Pricing

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Sustain use discount-->on-demand for computer resources
Committed use discount-->1 year or 3 year reservation-->sole tenant node
Preemptible VM or spot VM--> 60% discount up to 90% discount

Sustain use discounts

Sustained use discounts are applied automatically to usage within a project separately for each region, so there is no action required on your part to enable these discounts.

If you run a VM for more than 25% of a month you are eligible for discount between 20%-30%.

Discount is given on VCPU and memory usage.

For general purpose N2 and N2D predefined and custom machine types and for compute optimized machine types, you can receive a discount of up to 20%.

For general-purpose N1 predefined and custom machine types, sole-tenant nodes and GPU you can get a discount of up to 30%.

Limitations

Sustained use discounts automatically apply to VM created by GCE or GKE.
Sustained use discounts do not apply to VMs Created using the app Engine environment and dataflow.
Sustained use discounts do not apply to E2, A2 or Tau, T2D machine types

Committed use discounts

Ideal for workload with predictable resource needs.

When you purchase a committed use contract you purchase compute engine resources -such as VCPUS memory, GPU's Local SSD's, and sole-tenant nodes -at a discounted price in return for committing to paying for those resource for 1 year to 3 year.

Discount is up to 57% to 70%.

After purchase you are billed monthly whether you are using.
No upfront cost is required.

How it works?

For example if you purchase a commitment for 8 Cores and you run 24 cores during the month, you will receive committed use discount for 8 cores. The other 16 cores will be billed at standard, non-committed use rates.

Note: Keep in mind that after you purchase a set amount of commitments you are billed for those commitments monthly even if you don't use them.

You can't combine committed use discounts and sustained use discounts for the same resources.

Limitations

Projects that are in the free tier period with free credits and projects that do not have any payment history do not qualify for committed use discount.

Committed use discounts apply to usage in the project or cloud billing account in which those discount are purchased.

Committed use discounts do not apply to preemptible VM

If a reservation is attached to committed use discount, the reservation can't be deleted for the duration of the commitment.

Preemptible Instances

Preemptible instances are available at much lower prices -60% to 90% discount, compared to the price of standard VM's.

However google cloud might stop these instances if it needs to reclaim the compute capacity for allocation to other VM's.

Compute engine always stops preemptable instances after they run for 24 hours.

The GCP free Tier credits for compute engine do not apply to preemptible instances.

Preemptible instances cannot live migrate to a regular VM instances or Vice-Versa.

Not Covered under SLA.

GCP gives only 30 Sec time before reclaiming it.

Best suited for batch processing jobs of short durations.

Storage Options for Compute Engine

Zone persistent disk -->Standard PD-HDD, Balanced PD-SSD, SSD PD-SSD, Extreme PD-SSD

Note: Increase but not shrink the disk size.

Number of disks is dependent on VCPUs

Size of disk increase

Can't take snapshot of local disk

Block Storage type, NAS, Bootbales

Regional Persistent Disk -->Standard PD-HDD, Balanced PD-SSD, SSD PD-SSD, Extreme PD-SSD

Data is replicated between two zones in the same region block storage type Bootable disk.

Local SSD

Physically attached to host server, non-persistent, DAS, Not bootable.

Filestore

Shared storage for instance.

Cloud Storage Bucket-Object storage

Standard Storage

Nearline

Coldline

Archive

Create & Manage VM instance using Cloud shell

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Login to GCP console -->select your project -->Activate Cloud shell

Gcloud compute instances create myvm01 --machine-type n1-standard-2 --zone asia-south1-a

It will ask to authorize credentials, click on authorize.

Gcloud compute ssh myvm01 --zone=asia-south1-a

Gcloud compute images list

Gcloud compute instances create myvm02 --image-family xxxx --image-project xxxx --zone=us-central1-a

Gcloud compute instances describe myvm02

Gcloud compute instances stop/start/restart myvm01

Gcloud compute instances delete myvm01

Changing the Machine type of a vm instance

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If your existing machine is not a good fit for the workloads you run on your VM change the machine type of that VM (Vertical Scaling).

If your VM does not have a local SSD and is not part of a managed instance group (MIG), you can change the machine type of VM after stopping it.

If your VM has a local SSD, you cannot stop the VM to change its machine type unless you force it to stop, which causes you to lose all of the data on the local SSD.

Changing a machine type might also affect the sustain use discounts for that VM.

You can only change the machine type of a stopped VM. You cannot change the machine type of a running VM.

Take snapshot before updating size .

For VMs that do not have a local SSD and are not part of a MIG, you can change the machine type without affecting the following resources-

1. The VMs SSH keys
2. The VM configuration such as metadata
3. Persistent disk data including installed applications and app data.

Steps:

GUI

1. Go to Google Cloud console-->compute-->VM instances-->select the VM-->click edit
2. Now to modify the VM, stop it.
3. Go to edit--> Machine configuration -->select the machine type you want
4. Refresh console

CLI

1. Stop the VM first

`gcloud compute instances stop myvm01`

2. Change machine type.

`gcloud compute instances set-machine-type myvm01 --machine-type e2-standard-2`

3. For custom machine type you can use below command

`gcloud compute instances set-machine-type myvm01 --machine-type e2-custom-4-2048`

Resize your persistent disk of VM

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You can increase the size of your persistent disk when your vm instances requires additional storage space or high performance limit.

You can increase the size of PD but you cannot reduce.

VM using public images automatically resize the boot disk in all other cases you need to manually resize the boot disk or non-boot disks.

No need to stop the VM to resize.

Steps:

GUI

Login to GCP console-->compute engine-->create one VM with 10GB size of boot volume

Now, to modify this volume.

Go to disk -->select disk-->edit-->modify size =30 GB-->save.

Now check on VM details page, it shows 30 GB storage.

CLI via cloud shell

Gcloud compute disks resize diskname --size disksize

Now to resize in linux VM

Take ssh

Sudo su

Df -Th

Lsblk

Parted /dev/sda

Resize part

1

Yes

97G extend--> 100%

Quit

Now check again LSBLK

Images

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You can use O.S system to create boot disk for your instance. You can use one of the following image type -

Public Image-->

Public image provided and maintained by google, open source communities and third-party vendors, By default, all google cloud projects have access these images and can use them to create instances.

Custom Images--> Custom Images are available only to your cloud Project. You can create a custom image from boot disk and other images. Then use the custom images to create an instance.

Custom images are free of cost but do incur an image storage charges.

You can use custom images for following purpose-

Import a Virtual disk to compute Engine from your on-premises Environment or form VMs that are running on your local Workstation or an another cloud platform.

Create an image from bootdisk of your existing VM. Then use that to create New boot disk for your instance.

Copy one image to another image by using either the gcloud cli or the API you can also create an image from a custom image in a different project

Machine Image-->

A Machine image is a compute resource that stores all the configuration metadata, permissions and data from multiple disks of a VM instance you can use a machine image in many system maintenance, backup and recovery and instance cloning.

Machine image are the most ideal resources for the following cases-

1. Multiple disk backups.
2. Instance cloning.

Machine images are suitable for creating backup of all disks that are attached to a vm instances.

Machine images can be used to clone instances you can create multiple copies of an instances.

Information stored in a machine image:-

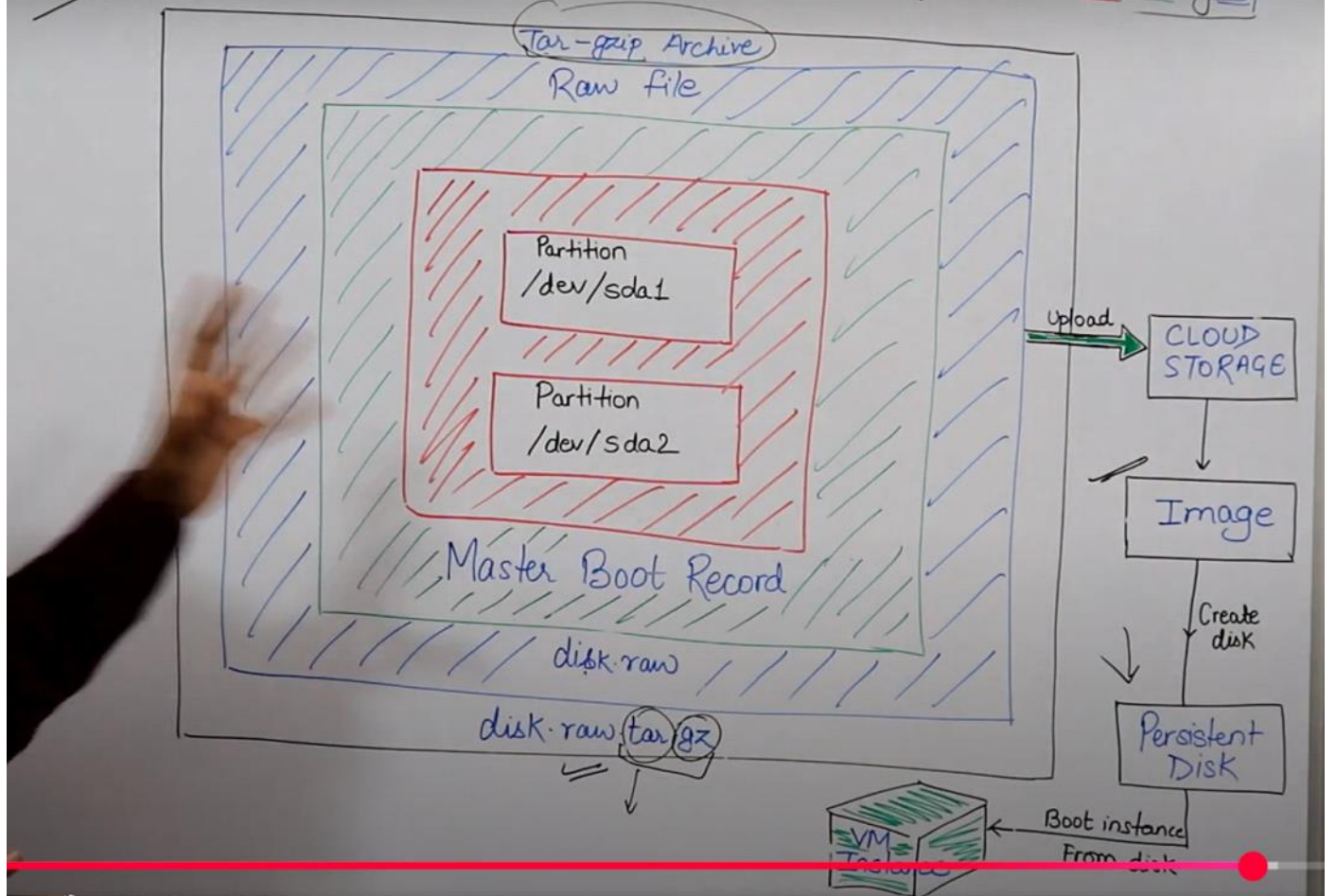
Description Machine type, labels, Network tags, maintain policy and instances metadata.

Machine image stored in cloud storage bucket.

Regardless of where the machine image is stored, you can still access the machine image globally.

Machine image is unchangeable but you can few properties.

Understanding Images, Machine Image and Custom Image | Lec-15 |



Machine Image and Custom Image

When to use which image ?

SCENARIOS	MACHINE IMAGE	PERSISTENT DISK SNAPSHOT	CUSTOM IMAGE	INSTANCE TEMPLATE
<u>Single Disk Backup</u>	YES	YES	YES	NO
<u>Multiple Disk Backup</u>	YES	NO	NO	NO
<u>Differential Backup</u>	YES	YES	NO	NO
<u>Instance cloning</u>	YES	NO	YES	YES
<u>Base image for Replication</u>	NO	NO	YES	NO

How to create Custom image and VM from it

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Go to GCP console-->compute-->VM instance-->create instance

Go to Compute-->VM instances-->image-->create Image.

Name-instance-1

Source-disk

Location-multi-regional{asia}

Keep instance running.

--> create

Go to compute-->create instance

Name region-asia-south1

Boot disk-->custom images-->select your image-->create instance.

Using CLI

Disable auto delete option for the disk

Gcloud compute instances set-disk-auto-delete VM-name --no-auto-delete --disk=sourcdiskname

Gcloud compute images create img-name --source-disk=source-disk --source-disk-zone+zone

Gcloud compute images describe image-name

Gcloud compute images delete image-name

Creating a machine image from a vm

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Go to gcp console-->create one vm instance-1

Now go to "machine instance"

Create machine image name-->mymachine -img source VM instance-->instance-1 Location->regional

-->create

Test

CLI using cloudshell

Gcloud compute machine-images create machine-image-name --source-instance=source vm-name --source-instance-zone=us-central1-a

Now create instance from machine image

Gcloud beta compute instances create vm-name --zone=zonename --source-machine-image=machine-image-name --machine-type=e2-standard-2

Cloud Storage

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Cloud Storage
Persistent Disk
File store

Google cloud database store
Bigtable
Filestore
Cloud sql
Cloud datastore
Cloud spanner
Memory store

Cloud storage

Cloud storage is a source for storing your objects in Google Cloud. An object is an immutable piece of data consisting of a file of any size and format.

You store objects in containers called buckets.

Ways to access/interact with cloud storage

1. console
2. Gsutil
3. Client libraries
4. REST APIs

Cloud storage classes

Standard storage
Nearline storage
Cold line storage
Archive storage

Cloud storage types and its characteristics

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Storage Classes

Standard storage

Nearline storage

Cold line storage

Achieve storage

A storage classes is a piece of metadata that is used by every object

The storage class set for an object affects the object availability and pricing model.

You can change the storage class of an existing object either by rewriting the object or by using object lifecycle management.

When you create a bucket you can specify a default storage class for the bucket if you don't specify then default storage is standard.

All storage classes provides the following-

Unlimited storage with no minimum object size

Worldwide accessibility and worldwide storage locations

Low latency

High durability 99.9999

Geo redundancy, if the data is stored in a multi- region or dual region

A uniform experience with cloud storage feature security tools and APIs

Object written to cloud storage must be redundantly stored in at least two different availability zones before the write is acknowledged as successful

Max size of an object stored in cloud bucket is 5 TB

There is a 100 ACL limit per object

A multiple upload can have upto 10,000 Parts. An individual part has a max size limit of 5 GB.

Object assembled from a multipart upload are subject to the overall 5TB size limit

Standard Storage

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Standard Storage is best for data that is frequently accessed (hot data) or stored for only brief periods of time.

When used in a region, standard storage is appropriate for storing data in the same location as GKE Clusters or compute engine instances that use the data.

Regional storage is redundant within a region.

When used in dual-region improved availability.

When used in multi-region ideal for string data that is accessed around the world, such as serving website content, streaming videos, executing interactive workloads or serving gaming data or apps.

Multi-regional is geo-redundant stored across 2 or more regions which are 500 miles apart.

Availability of standard storage data is

Location type	Availability SLA
Multi region	99.95%
Dual Region	99.95%
Regional	99.90%

Nearline Storage

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Nearline storage is a low-cost, highly durable service for storing infrequently accessed data.

Nearline storage is better choice than standard storage in scenarios where slightly lower availability, a 30-day minimum storage duration and cost for data access are acceptable trade-offs for lowered at-rest storage costs.

Nearline storage is ideal for data you plan to read or modify on average once per month.

Availability of nearline storage data is

Location Type	Availability SLA
Multi region	99.9
Dual Region	99.9
Regional	99.0

Cold line storage

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Cold line storage provides very low cost highly durable storage service for storing infrequently accessed data (cold data).

Cold line storage is ideal for data you plan to read or modify at most once a quarter.

Availability is same as nearline

Archive Storage

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Archive Storage is the cheapest highly durable storage service for data archiving online backup and disaster recovery.

Unlike the coldest storage service offered by other cloud providers, your data is available within milliseconds, not hours or days.

Archive storage has higher cost for data access, as well as 365 days min storage duration.

Archive storage is the best choice for data you plan to access less than once a year.

Availability is same as nearline or cold line.

To maximize the availability of your data

Consider storing your data in a multi-region or dual region bucket location if high availability is a tip requirement.

All data is stored geo-redundantly in these locations, which means your data is stored in at least two geographic separated regions. IN the unlikely event like region wide outage, buckets in geo-redundant locations remains available, with no need to change storage path.

You cannot change bucket location or project after creations.

Cloud storage pricing, Bucket & Object

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Cloud storage pricing is based on the following components:-

1. Data storage
2. Data processing
3. Network usage

Data storage:

The amount of data stored in your buckets storage rates vary depending on the storage class of your data and locations of your buckets.

Data processing:

The processing done by cloud storage, which includes operations charges any applicable retrieval fees and inter-region replication.

Network Usage:

The amount of data read from or moved between your buckets

For eg data storage cost in Iowa.

	Standard	Nearline	Cold line	Archive
Regional	\$0.020	\$0.010	\$0.004	\$0.0012
Dual-regional	\$0.022	\$0.011	\$0.0044	\$0.0044
Multi-regional	\$0.026	\$0.010	\$0.007	\$0.004

Note:

Dual regions are billed to both underlying regions at the above prices.

For eg- Standard storage in a dual-region compressed of IOWA and Oregon will be billed at \$0.22 per GB per month for IOWA and \$0.22 per month for Oregon.

Data processing fees-

Data processing costs consist of operation charges, which apply to all request mode to cloud storage, retrieval fees, which apply to reading data stored in certain storage classes & Inter-region replication charges, which apply to data written to dual regions & multi regions.

Price is \$0.05/10,000 Operations in nearline & cold line

Retrieval/Network usage fees-

Applies when you read, copy, move or rewrite object data or metadata that is stored using nearline storage, cold line storage or Archive storage.

Standard Nearline Coldline Archive
0/GB 0.02GB 0.02GB 0.005GB

Default inter-region replication is free turbo replication is 0.04GB chargerable

Data move within the same location is free

Data moves different locations on the same continent is chargable

Data moves between different continents in chargable.

Bucket & Objects; Turbo Replication

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Bucket:

Bucket is like a folder for your objects

Using buckets you can organize and maintain access permissions.

You cannot create nested buckets

Bucket has name, default storage class, geographic locations

Name and location can only be changed by deleting and recreating.

There is only one google cloud storage namespace, which means every bucket must have a unique name across the entire google cloud storage namespace.

Object names must be unique only within bucket.

Object:-

Object is file that you want to store.

Can be up to 5TB

Object has data & Metadata

No limit on number of objects in buckets.

Objects are immutable and you can overwrite objects.

Turbo Replication:

Turbo replication is a cloud storage premium feature designed to asynchronously replicate newly written cloud storage objects associated with any insert, rewrite, copy or compose operation-regardless of object size to a separate region within a target of 15 minutes

Create a bucket & upload object

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Login to GCP console-->go to cloud storage --> Bucket-->create bucket.

1. Name your bucket labels (optional)
2. Choose where to store your data
 - Multi region
 - Dual region
 - Region
3. Choose a default storage class for your data.
 - Standard
 - Nearline
 - Cold line
 - Archive
4. Choose how to control access to object
 - Uniform
 - Fine grained
5. Choose how to protect object data protection tools.
 - None
 - Object versioning
 - Retention policy
6. Data Encryption
 - Google Managed Encryption Key
 - Customer Managed Encryption Key

CLI through Cloudshell

Open cloudshell-->select your project

Ls -a

Sudo rm -rF \$Home -->Restart cloudshell

1. Create a bucket
Gsuil mb -b on -l us-east1 gs://tgbucket454/
2. Upload an object into your bucket
Cloudshell--> : -->Upload-->ofile-->upload file

Gsuil cp bhupeder.jpg gs://tgbucket454

3. Download the object from your bucket.

Gsuil cp gs://tgbucket454/bjupender.jpg //home/aws3may/download.jpg

4. Copy the object to a folder in the bucket

Gsuil cp gs://tgbucket454/bhp.jpg gs://tgbucket454/myfolder/newjpg

5. List content of a bucket or folder

Gsutil ls gs://tgbucket454

6. List details for an object

Gsutil ls -l gs://tgbucket454/bhup.jpg

7. Make object publicly accessible

Gsutil iam ch allusers:objectViewer gs://tgbucket454

Revert permissions

Gsutil iam ch -d allusers:objectViewer gs://tgbucket454

8. Delete an object or bucket

Gsutil rm gs://tgbucket454/bhup.jpg

9. Delete entire bucket and its object

Gsutil rm -r gs://tgbucket454

10. Object versioning on bucket

Gsutil versioning set off/on gs://tgbucket454

11. Get list of bucket object

Gsutil ls -a gs://tgbucket454

12. Updating version

Gsutil cp <source> <destination>

Object versioning & Lifecycle Rules on Buckets

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To support the retrieval of objects that are deleted or replaced, cloud storage offers the object versioning features

Once Enabled object versioning--

Cloud storage retains a non-current object version. Each time you replace or delete alive object version as long as you do not specify the generation number of the live version.

Non-current versions retains the name of the object, but are uniquely identified by their generation number.

Non-current versions only appear in request that explicitly call for them to be included.

If you disable object versioning--

The bucket no longer accumulates new Non-current version of object.

Object versions that already exist in the bucket are unaffected.

Object Lifecycle Management:

To support common use cases like setting a TTL for object, retaining non-current versions of objects to help manage cloud storage offers the object lifecycle management features.

In Order to use object lifecycle management, you define a lifecycle configuration, which much be set on a bucket. The configuration contains a set of rules which apply to current and future objects in the bucket when object meeting the criteria of one of the rules cloud storage automatically perform on the object here are some examples use cases--

Downgrade the storage class of objects older than 365 drops to cold line storage

Delete object created before 1 jan 2022

Keeps only the 3 most recent versions of each object in a bucket with versioning enabled.

Object Lifecycle configuration

Each lifecycle management configuration contains a set of rules each rule contains one actions and one or more conditions.

An object has to match all of the conditions specified in a rule for the action in the rule to be taken.

If you specify multiple rules that contains the some action, the action is taken on an object when that object matches the conditions in any of the rules.

If multiple rules have their conditions simultaneously for a single object, cloud storage performs the

action associated with only one of the rules, based on the following consideration.

The deleted action takes precedence over any set storage class action.

The set storage class action that switches the object to the storage class with the lowest at-rest storage pricing takes precedence.

Signed URL

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How to create Signed url?

A signed url is a url that provides limited permissions and time to make a request.

Signed URL's contains authentication information in their query string allowing users without credentials to perform specific actions on a resource

Key parameters in signed URL

1. X-Goog-Algorithm-->The algorithm used to sign the URL.
2. X-goog-Credentials-->Information about the Credentials used to create the signed URL.
3. X-Goog-date-->The date and time the signed URL became usable, YYYYMMDD 'T' HHMMSS 'Z'
4. X-Goog-Expires-->The length of time the signed URL remained valid measured in Sec. In our video we will take 60 Sec. Max duration is 60,4800 Sec i.e., 7 Days.
5. X-Goog-signheader-->Header that had to be included as part of any request that used the signed URL.
6. X-Goog-signature-->The authentication string that allowed request using this signed URL to access Resources.

Static Website Hosting on Cloud Storage

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Login to GCP Console-->Create a Bucket
Go to cloudshell

Wget "pastlink"

Ls

O/P samplestyle_banner.zip

Now create one directory and do all things in it.

-->Mkdirwebsite

-->MV samplestyle_banner.zip Website/

-->cd website/

-->Unzip samplestyle_banner.zip

-->gsutil -m cp -r sample style_banner gs://bucket5543

Go to console-->bucket->folder->select all files or object->permissions->Add principal-Allusers select
role->cloud storage->viewer

Now test your static website copy URL of index.html file
And check in browser.

VPC Network

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Virtual private cloud Network is just like your physical network except that it is virtualized within google cloud.

A VPC network is a global resource that consist of a list regional virtual subnetworks in Data centers, all connected by a global wide Area network.

A VPC Network provides-

1. Provides connectivity for your VM, including GKE & app engine instances.
2. Offers native internal TCP/UDP load balancing.
3. Connects to on-premises network using cloud VPN tunnels and cloud Interconnect attachments.
4. Distributes traffic from Google cloud external Load balancers to backends.
5. VPC always lies inside a project one project can contain multiple VPC network.
6. New project start with a default VPC that has one subnet in each region.

VPC Networks have following properties-

VPC networks, including their associated routes and firewall rules are global resources. They are not associated with any particular region or zone.

Subnets are regional. Each subnet is associated with a region you can create more than one subnet per region.

Each subnet is having unique network range inside a project.

Inside(Ingress) and outside(Egress) traffic can be controlled by firewall rules.

Resources within a VPC, can communicate with one another by using Ipv4 and IPv6 address.

An Organization can use shared VPC to keep a VPC network in a Common host project.

VPC networks can be connected to other VPC networks in different Projects or Organizations by using VPC network peering.

VPC Network do not support broadcast or multicast address within the network.

Auto mode VPC networks create subnets in each region automatically.

You cannot connect directly Auto mode VPC network to one another because they uses same ranges

Create Automatic VPC using console & Cloud shell

Friday, April 18, 2025 8:52 PM

1. Login to GCP console -->VPC Network-->create VPC network

Name -my-auto-vpc ULA IPv6-->disabled.

Subnet--> Custom
Auto

IPv4 Firewall Rules--> TCP 22

Dynamic Routing mode --> Regional
Global

MTU(maximum Transmission unit) -->1460/1500/8896

-->Create

2. Go to compute Engine-->Create two VM's in different region
For Eg: US-Central1 7 & Asia-south1

3. Take ssh of both VM's and ping each other with their private IP.

MTU is the size in bytes or the largest packet supported by a network layer protocol including both headers & Data

Use Higher MTU (>1600) can be used only if the source & Destination interfaces are in the same subnet and uses private IP's.

Create Custom VPC using GUI and CLI

Friday, April 18, 2025 9:25 PM

1. Login to GCP console -->VPC Network-->create VPC network

Name -my-auto-vpc ULA IPv6-->disabled.

Subnet--> Custom
Auto

IPv4 Firewall Rules--> TCP 22

Add subnet

Name-Subnet -us-east1
IPv4 range-10.0.2.0/24
Done

IPv4 firewall rules
Allow-icmp
Allow-ssh
Allow-RDP

Dynamic Routing mode --> Regional
Global

MTU(maximum Transmission unit) -->1460/1500/8896

-->Create

VPC Network peering

Friday, April 18, 2025 10:00 PM

Google Cloud network peering allows Internal IP address connectivity across two VPC networks regardless of whether they belong to the same project/organization or not.

In VPC network peering, traffic stays within Google's network and doesn't traverse the public internet.

Key Features

VPC network peering works with Compute Engine GKE and App Engine flexible environment.

Each side of a peering association is set up independently. Peering will be active only when the configuration from both sides matches.

Either side can choose to delete the peering association at any time.

VPC network peering supports IPv4 connectivity only.

CIDR range cannot overlap in two peered networks.

Transitive peering is not supported.

One VPC can be directly peered with a maximum of 25 VPCs.

Peering is allowed with shared VPC.

Cloud NAT Gateway

Friday, April 18, 2025

10:34 PM

Network Address Translation(NAT) is a method of remapping one IP address space into another by modifying network address information in the IP header of packet usually this is done in NAT device.

IP masquerading or NAT is a technique that hides an entire IP address, Usually the private IP addresses, behind a single Public IP.

The hidden addresses are changed into a single public IP as the source of the outgoing IP packet so they appear as originating from routing devices and not from the hidden cost.

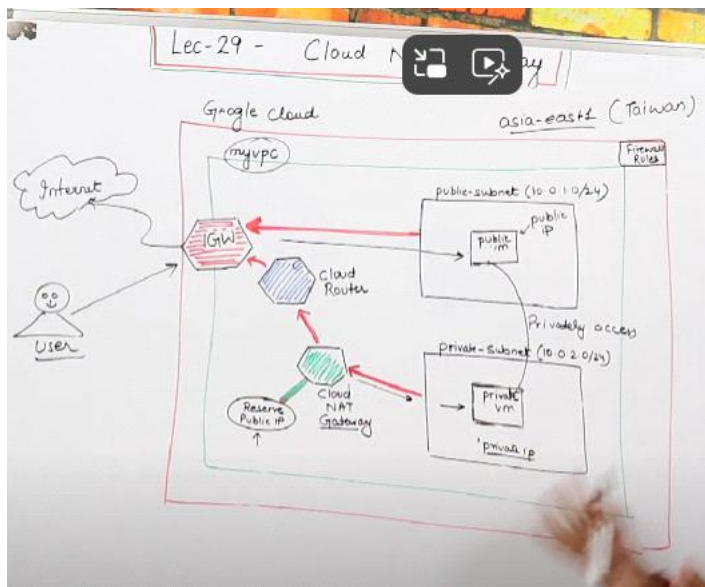
Cloud NAT allows outbound traffic but restrict Inbound traffic from external sources

Cloud NAT allows your resources to create Outbound connects to the Internet without giving public IP to them.

You can have multiple cloud NAT gateways in the same region of a VPC but each gateway is configured for different subnet.

Cloud NAT gateway provides outgoing connectivity for the following resources-

1. Compute Engine VM instances without external IP address.
2. Private GKE clusters.
3. Cloud Run instances through serverless VPC access.
4. Cloud function instances through serverless VPC access.
5. App Engine standard environment instances through Serverless VPC access.
6. Cloud NAT gateway cannot apply to any custom route whose next hop is not the default internet gateway.
7. Google Cloud NAT is based on Andromeda i.e., google network virtualization stack.
8. There is no intermediate NAT proxies in the data path Each VM is programmed by GCP to NAT using assign ports.
9. Each cloud NAT gateway is associated with single VPC network region and cloud routers cannot use same cloud NAT in two or more regions.
10. You cannot use NAT for communication between VM's in a VPC Network. IT is designed to provide connectivity to the internet.



Cloud NAT Pricing

Cloud NAT pricing is based on following usage-

1. A hourly price of NAT gateway that is based on the number of VM.
2. Per GB cost for Ingress or egress data that is processed by the gateway.

No of assigned VM Price per hour Price per GB processed(ingress/egress)

Upto 32 VM \$0.014xno of VMx hours \$0.0

Above 32 VM \$0.44 \$0.045

Load Balancers

Saturday, April 19, 2025 9:05 AM

A load balancer distributes user traffic across multiple instances of your application. By spreading the load, load balancing reduces the risk that your applications experience performance issues.

Cloud load balancer is built on the same frontend-serving infrastructure that powers google it supports 1 million+ queries per second with consistent high performance and low latency traffic enters cloud load balancing through 80+ distinct global load balancing locations, maximizing the distance traveled on Google's fast private n/w backbone.

Google cloud offers the following load balancing features:-

1. Single any cast Ip address.
2. Software-defined load balancing.
3. Seamless autoscaling no pre warming.
4. Layer-4 and layer 7 LB.
5. External and internal LB.
6. Global and regional LB.
7. Enable cloud CDN for HTTP(s) LB to optimize application latency for your users with a single checkbox.
8. Manage SSL certificate and decryption.

Load Balancer can be opted on the basis of your requirement. In google you have various aspects:-

1. External versus Internal LB
2. Global versus regional
3. Premium v/s standard tier
4. Proxy v/s pass-through
5. Traffic type
6. DDoS(Distributed denial of service) protection

Types of LB

Saturday, April 19, 2025 11:08 AM

Global LB when your backends are distributed across multiple regions.

Regional LB when your backends are in one region or when you have Jurisdictional compliance requirements for traffic to stay in a particular region.

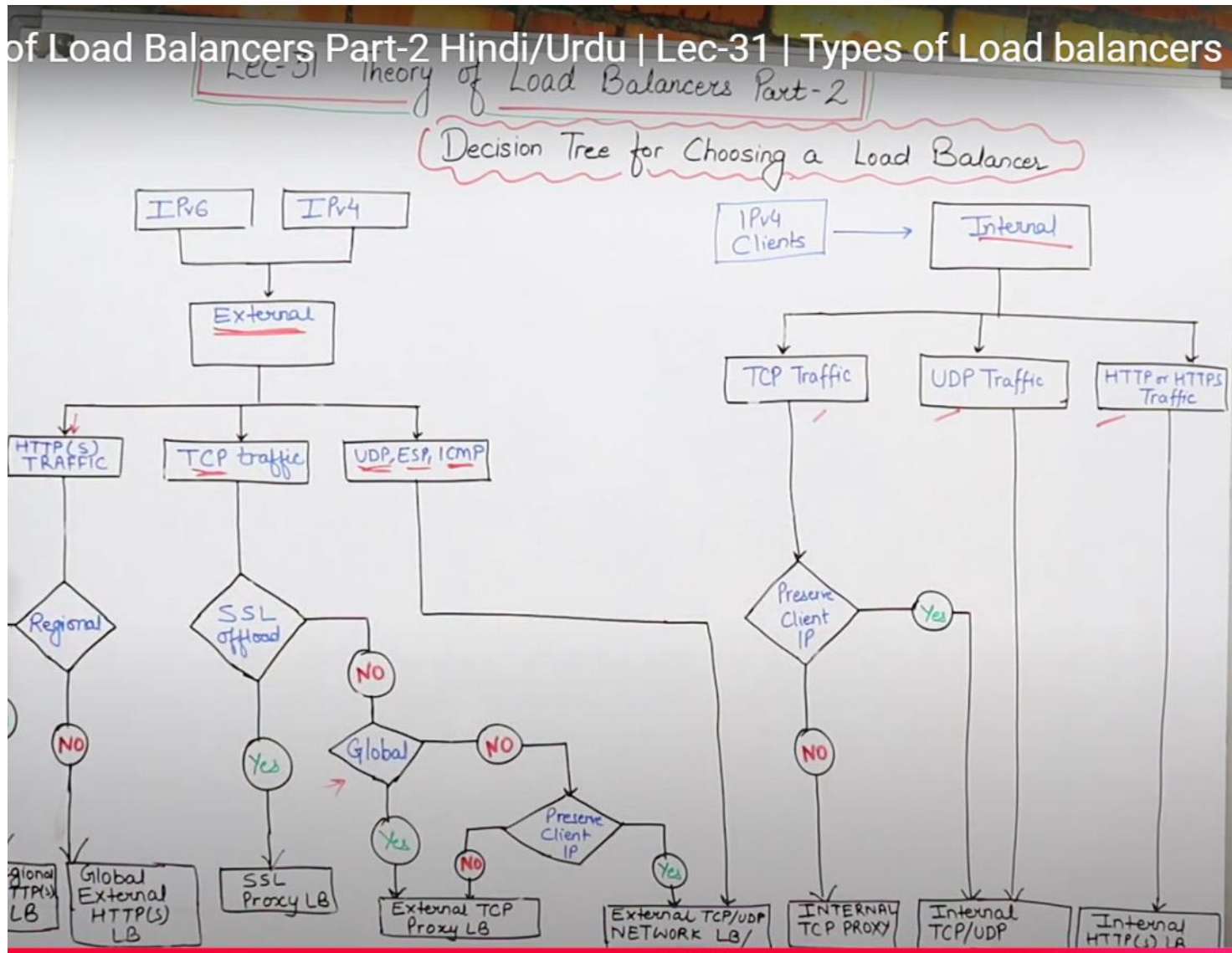
External Load balancers distributed traffic coming from the internet to your google cloud VPC.

Premium tier delivers traffic on googles premium backbone. It gives high performance and low latency.

Standard tier uses regular ISP network. It is low cost alternative for apps that don't have strict req for latency & performance.

Proxy load balancers terminate incoming client connections and open new connection from the LB to the backends.

Pass through LB do not terminate client connections. Instead, load-balanced packets are received by backend VMs with the packet sources, destination and if applicable, port information. Unchanged connections are terminated by the backend VMs responses from the backend VMs go directly to the clients not back through the load balancer. The term for this is direct server return.



Underlying technology for google cloud load balancers-

1. Google front ends(GFE) are software defined distributed systems that are located in google points of presence(PoPs) and perform global load balancing in conjunction with other systems and central plane.
2. Andromeda is google cloud SDN virtualization Stack
3. Maglev is a distributed system for network LB.
4. Envoy proxy is an open source edge and service proxy, designed for cloud native applications.

Autoscaling

Sunday, April 20, 2025 7:18 PM

Autoscaling

1. Vertical Scaling(Manual) Increasing SKU
2. Horizontal Scaling(Automatic)Add on/offload new servers.

IAM

Sunday, April 20, 2025 7:30 PM

Identity and Access Management

IAM lets you grant granular access to specific google cloud resources and helps prevent access to other resources.

IAM lets you adopts the security principle of least privilege, which states that nobody should have more permissions than they actually need.

With IAM you manage access control by defining who(Identity) has what access(role) for which resources.

IAM permission to access a resource is not granted directly to the end user Instead permissions are group into roles and roles are granted to authenticated principals(members)

Policy defines and enforces what roles are granted to which principals. Each allow policy is attached to a resource.

When an authenticated principal attempts to access a resource, IAM checks the resource's allow policy to determine whether the action is permitted.

Policy-->Member+role

Features of IAM

Sunday, April 20, 2025 7:49 PM

Lets you authorize who can take specific actions on resources to give you full control and visibility on your google cloud services centrally.

Permissions are represented in the form of service resource verb.

Can map job functions into groups and roles.

With IAM, users only get access to what they need to get the jo done.

Cloud IAM enables you to grant access to cloud resources at fine-grained levels, well beyond project level access.

Basically IAM has three important parts-

- A. Principal/members- A principal can be a
 - 1. Google Account
 - 2. Service account
 - 3. Google group
 - 4. Google workspace
 - 5. Cloud Identity domain
 - 6. Authenticated users
 - 7. All Users
- B. Role:- A role is a collection of permissions. Permissions determine what operations are allowed on a resource when you grant a role to a principle, you grant all the permissions that the role contains.
- C. Policy- Policy is a collection of role bindings that bind one or more principals to individual roles when you want to define who

IAM

Sunday, April 20, 2025 9:33 PM

In IAM you grant access to principals. Principals can be of the following types:

A. Principal/members- A principal can be a

1. Google Account
 2. Service account
 3. Google group
 4. Google workspace
 5. Cloud Identity domain
 6. Authenticated users
 7. All Users
-
1. Google Account - Any email address that's associated with a google account can be an Identity, including gmail.com or other domains.
 2. Service account- A service account is an account for an application or compute workload instead of an individual end user.
 3. Google group- is a collection of google accounts and service accounts every google group has a unique email address that's associated with the group.
You can grant and change access controls for a whole group at once instead of granting or changing access controls once at a time for individual users or service accounts.
 4. Google workspace account- Google workspace accounts are associated with your organization internet domain name, such as example.com when you create a google account for a new user, such as username@example.com, that google account is added to the virtual group for your google workspace account.
 5. Cloud Identity domain- A cloud Identity domain is like a google workspace account, because it represents a virtual group of all google accounts in an organization. However cloud identity domain users don't have access to google workspace applications and features.
 6. All Authenticated users- The value all Authenticated users is a special identifier that represents all service accounts & all users on the internet who have authenticated with a google account. This identifier includes accounts that are not connected to google workspace accounts or cloud identity domain, such as personal Gmail accounts users who are not authenticated such as anonymous visitors are not included.
 7. All users- the value all users is a special identifier that represents anyone who is on the internet, including authenticated and unauthorized users.

Roles

A role contains a set of permissions that allows you to perform specific actions on google cloud resources.

You don't directly grant users permissions in IAM. Instead you grant them roles, which bundle one or more permissions.

To make permissions available to members including users, group and service accounts you grant roles to the members.

There are three types of roles in google cloud IAM-

Basic
Predefined
Custom role

Basic-

Owner -

Super user at project level
Add or remove members
Delete & create projects
Setup billing for projects
Editor -

Modify codes
Deploy add
Configure services
Stop & Start service

Viewer -

Read only access

Billing Admin -

Manage billing
Add or remove administrators.

Browser -

No access to cloud storage

Predefined roles-

Provides granular access for a specific service and is managed and defined by google cloud.

Prevents unwanted access to others resources

Google is responsible for updating and adding permissions as necessary

You can grant multiple roles to the same users

Custom roles-

Provide granular access according to a user-defined list of permissions.

You can create a custom IAM role with one or more permissions and then grant that custom role to

users or groups.

Custom roles are not maintained by google.

You can grant multiple role to a user or a group.

Policy-

A policy is a collection of bindings audit configuration and meta data.

A binding associates one or more members with a single role and any context-specific conditions that change how and when the role is granted.

Each binding includes the following fields.

A member, known as an identity or principal can be

- User account
- service account
- Google group
- Domain

A role which is named collection of permissions that grant access to perform actions on google cloud resources.

A condition which is a logical expression that further constraints the role binding based on attributes about the request, such as its origin the target resource etc.

Definition

You can grant roles to users by creating an allow policy which is a collection of statements that define who has what type of access.

IAM - Service Account

Monday, April 21, 2025 9:42 AM

A service account is a special kind of account used by an application or a virtual machine (vm), not a person.

Application uses service accounts to make authorized API calls, authorized as either.

- The service account itself
- As google workspace
- As cloud identity users through domain-wide delegations.

A service account is identified by its email address, which is unique to the account.

Saname@projectid.iam.gservice.com

Each service account is associated with two sets of public/private RSA key pairs used to authenticate to google.

Types of service accounts:-

1. Default service account.
2. User-managed service accounts.

A service account is also a resource with IAM policies attached to it, which means you can define who can use the account and who can perform specific actions on the service account.

IAM Service Account

Monday, April 21, 2025 10:05 AM

A service Account is a special kind of account used by an application or compute workload, such as a compute engine VM instance, rather than a person.

For Eg-> A service account can be attached to a compute engine VM, so that application running on that VM authenticate as the service account. In addition the service account can be granted iam roles that let it access resources.

Difference between Service account & user account-

- Service account do not have passwords and cannot log in via browsers or cockes
- Service accounts are associated with Public/private RSA key pair that are used for authentication to google and for siging data
- You can let other users or service account impersonate a service account.
- Service account do not belongs to your google workspace domain, unlike user account.

Types of Service accounts

- Default service account
- User managed service account
- Google managed service account

Default service account - when you enable or use some google cloud services they create default service accounts that enable the service to deploy jobs that access other google cloud resources. These accounts are known as default service account.

There two services creates service account App Engine, compute engine and any google cloud service that uses app engine or compute engine

User managed service account - You can create user-managed Service account in your project using IAM API, console and CLI.

By default you can create 100 user managed Service account in a project

When you create user managed service account in you project, you choose a name for the service account which uses the following format

Service account name@project-id.iam.gserviceaccount.com

Google managed service account-

Some google cloud services needs access to your resources so that they can act on your resources so that they can act on your behalf for eg, when you use cloud run to run a container the service needs access to any pub/sub topics that can triggers the container.

To meet this need google create google managed service account not listed in the service page.

IAM Service account key types

Monday, April 21, 2025 10:54 AM

Types of Keys for Service Account

Google Managed keys pair

User Managed Key pair

Google managed key pair are used by service account credentials API and by google cloud services such as App Engine & Compute Engine.

User Managed Key pair

You can create user-managed key pairs for a service account, then use the private key from each key pair to authenticate with google API's. This private Key is known as a service account key. Each service account can have up to 10 Service account Keys.

User is responsible for the security of private key.

Identity Aware Proxy

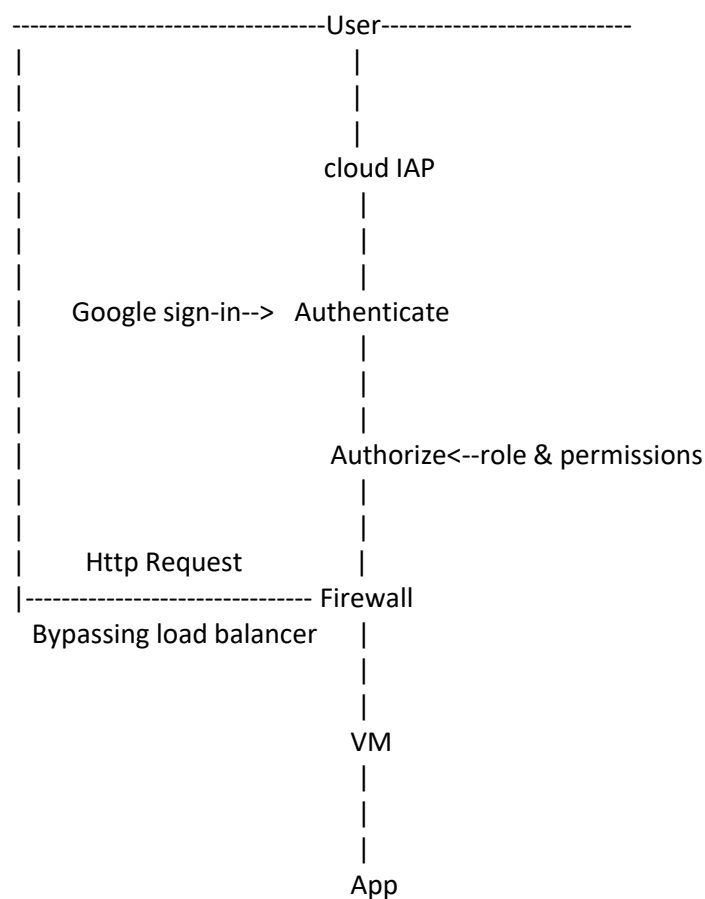
Monday, April 21, 2025 12:03 PM

IAP lets you establish a central authorization layer for applications accessed by HTTPS, so you can use application-level access control model instead of relying on network-level firewalls.

How IAP works?

When an application or resource is protected by IAP, It can only be accessed through the proxy by principals also known as users who have the correct identity and access management role.

When a user tries to access an IAP-Secured resource, IAP performs authentication and authorization checks.



- IAP does not support or use VPN
- IAP allows access to webapps from anywhere
- IAP allows the zero trust security model
- IAP supports google workspace account
- IAP works for app engine, cloud run, compute engine, GKE and on premises

App Engine

Monday, April 21, 2025

9:03 PM

The App engine is Google's PaaS offering. It's a compute service that provides a managed platform for running applications.

As this is a managed service, your focus should be on the application only & google will manage the sources needed to run the application.

It was GCP earliest service introduced in 2008.

Applications in app engine across multiple servers.

No usage charges - pay for resources provisioned.

Features of App Engine:-

App Engine makes deployment maintenance and scalability easy - so that the users may focus on innovation.

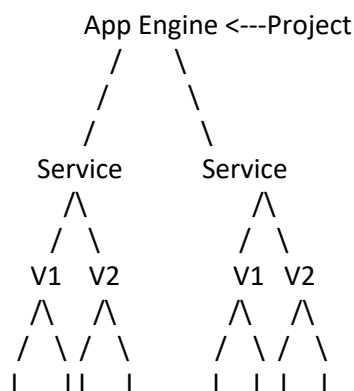
Automatic load balancing & scaling.

Application Versioning and splitting.

Supports several programming language like go, python, node js, java, ruby.

Suitable for web applications.

Services and versions



V=Version

I=Instance

Features	Standard	Flexible
Instance Startup	Seconds	Minutes
SSH Access	No	Yes

Write to local disk	No(some runtimes have read and write access to the /temp/directory Yes, Ephemeral (disk, initialized on each VM startup) can access app engine resources
---------------------	---

Need of VPC	No	Yes
-------------	----	-----

Pricing Model	After free daily use, pay per instances class, with automatic shutdown Pay for resource allocation per Hour.
---------------	---

Scaling to Zero	Yes	Min 1 Instance
-----------------	-----	----------------

Scaling	Manual, basic, Automatic	Manual & Automatic
---------	--------------------------	--------------------

Sandbox Environment App Run in docker container, support more language.

Difference Between compute Engine & App Engine

	compute Engine	App Engine
Service model	IAAS	PAAS
Type of Service	unmanaged	Managed by google
control and Flexibility over resources computing resources		Control Over Resources More Less control over
Cost	Less	High
Running Instance	Min 1	Scale to Zero
AutoScaling	slow scaling	fast scaling

App Engine - Flexible

The Flexible Environment provides more options and control to the developers who want to use App Engine, without the language constraints the standard environment.

Docker containers as basic building blocks. Can be auto-scaled.

Cloud Run

Monday, April 21, 2025 11:55 PM

Cloud Run is managed compute platform that lets you run containers directly on top of googles scalable Infrastructure.

Launched in 2019

AWS fargate and Azure container instances are equivalent to cloud run.

You can deploy code written in any programming language.

Containers can be called through web requests or pub/sub Events.

Cloud run offers Automatic scaling

Cloud run provides for an open API & runtime environment.

Cloud run comes with a free-tier with additional billing done on a pay per use basis.

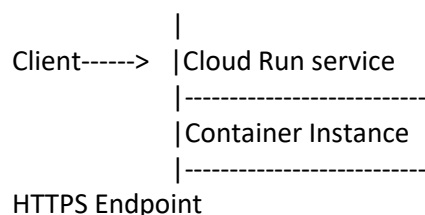
Its serverless and runs stateless containers.

Two way to run your code-

On cloud run, your code can either run continuously as a service or as a job.

Cloud Run services - used to run code that responds to web requests or events.

Cloud run jobs - Used to run code that performs work (a Job) and quits when the work is done.



A cloud run service provides you with the infrastructure required to run a reliable HTTPS endpoint you responsibility is to make sure your code listens on a TCP port and handles HTTP requests.

Features

1. Unique HTTP endpoint for every service every cloud run service is porivded with an HTTPS endpoint on a unique subdomain of the run.appdomain and you can configure custom domains as well.
2. Fast request-based autoscaling
3. Built-in traffic management --->Every deployment creates a new immutable revision. You can route incoming traffic to the latest version, roll-back to a previous version, or split traffic to multiple revisions at the same time.
4. Private and public services.

There are two pricing models.

1. Request based - If a container instance is not processing requests, the CPU is not allocated and you are not charged.
2. Instance based - You are charged for the entire lifetime of your container instance and the CPU is always allocated. There is no per-request fee.

Cloud Function

Tuesday, April 22, 2025 11:58 PM

With cloud functions you write simple single-purpose functions that are attached to Events emitted from your cloud infrastructure and services.

A Pay as you go function as a service (FaaS) to run your code with Zero server management.

Cloud functions can be written using:

- a. Node.js
- b. Python
- c. Go
- d. Java

Automatically scales based on load without thinking about the infrastructure.

Built-in Security at role and per function level based on the least privileged principle.

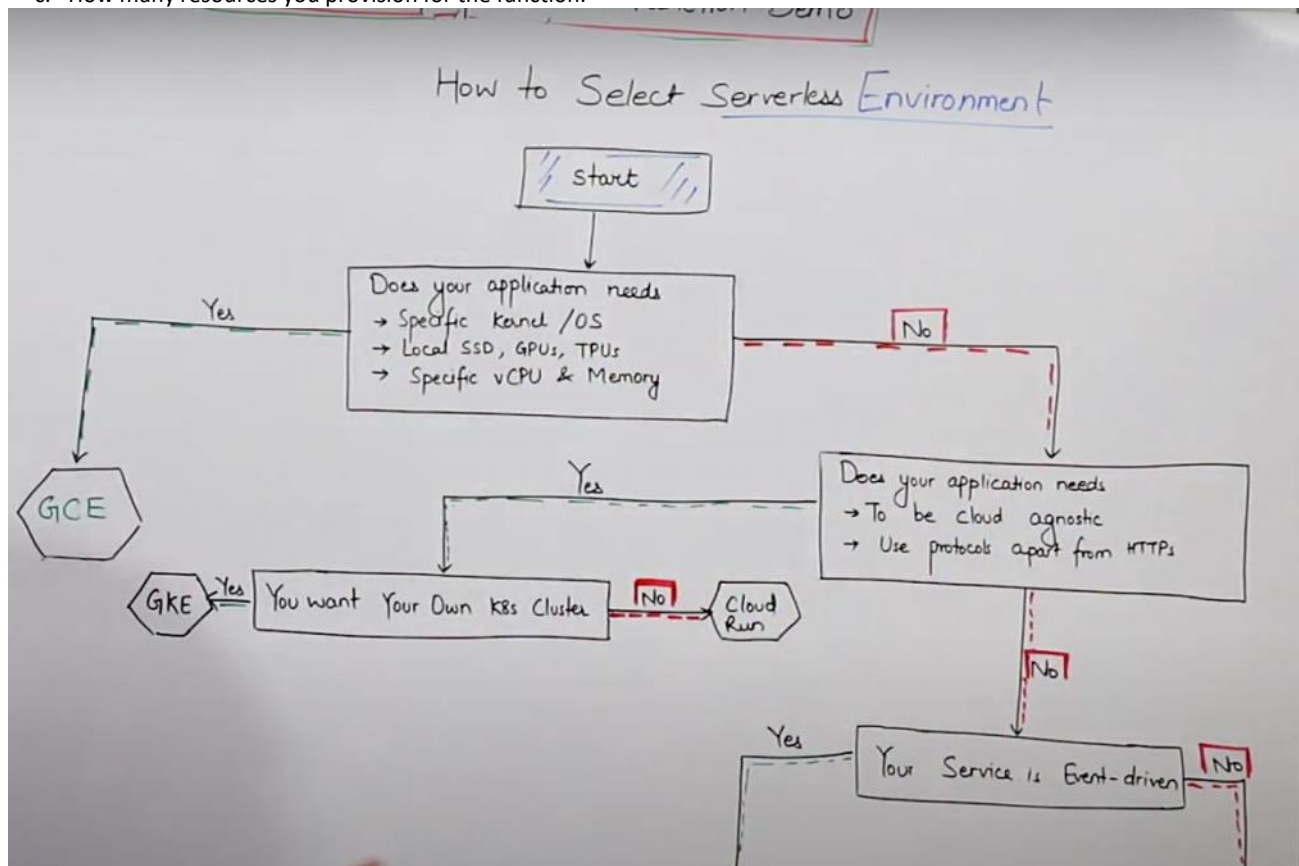
Allow you to trigger your code from Google cloud firebase & Google assistant or call it directly from any web, mobile or backend application via HTTP

To act on events, you shall define a trigger binding a function to a trigger enables you to act on events. The most common triggers are

- 1. Google cloud storage.
- 2. Pub/sub Event.
- 3. HTTP

Pricing = GCP pricing account to

- a. How long your functions runs
- b. How many times it is invoked.
- c. How many resources you provision for the function.





Google Cloud Function Demo

Differences b/w GAE, GCF, GCR

	App Engine	Cloud Functions	Cloud Run	Cloud Run For Anthos
Deployment type	App	Function	Container	Containers
Scaling	Yes	Yes	Yes	Only using Pods
Runtime Support	Java, Node.js, Python, Go, PHP	Node.js, Python, Go, Java	All Available	All Available
Access Control	Auth 2.0, CICP, Firebase Auth, Google Sign-in	IAM Permissions	IAM Permissions, CICP, Google Sign-in, Firebase Auth	Cluster-Only, VPC-Only
HTTP/gRPC	No	No	No	Yes
Custom domains	Yes ✓	No	Yes ✓	Yes
GPU/TPU Support	No.	No -	No	Yes
Timeout	1 min ✓	9 min.	15 min. ✓	15 min.
VPC networking	No standard Yes → Flex	Yes	No	Yes

Cloud SQL

Thursday, April 24, 2025 1:07 PM

Cloud SQL is fully managed relational database service for MySQL, Postgres, and SQL server

Cloud SQL is fully hosted on Linux

Cloud SQL is a web service that allows you to create, configure & use relational databases that live in google cloud.

Any existing databases in MySQL, Postgres or any other structured database can be migrated to cloud SQL.

Google handles replications, patch management and database management and database managed to ensure availability and performance.

Fully managed MySQL (5.6, 5.7 & 8.0)
Postgres SQL (9.6,10, 11, 12, 13, 14)
SQL Server (2017, 2019)

Easily Scale up to 96 vCPU of RAM(vertical Scalable)
And 624 GB storage available

Up to 64 TB of storage available

Support for private IP

Data replication between multiple zones with automatic failover.

Import and export databases using MySQL dump or import and export CSV files.

Instance cloning

To use cloud SQL effectively you should understand some of the concepts on which it is built.

1. Cloud SQL Instance --> A cloud SQL instance corresponds to one VM. The VM includes the databases instance and accompanying software containers to keep the database instance up and running.
2. Database Instance --> A database instance is the set of software and files that operate the databases, MySQL, Postgres SQL or SQL Server.
3. Public & Private Ip --> In cloud SQL Public ip means that the instance is accessible through the public internet, but are accessible through a VPC cloud SQL instances can have both a public and a private IP.
4. High Availability --> HA in cloud SQL works by having two sync instances : primary and standby. Each instance has exactly one VM. Each instance is in a different zone in the same region.
5. Failover --> A failover is when cloud SQL switches from the Original primary instances to the standby instance.

6. Standby Instance --> The standby instance is used in HA to replace the primary instance when failover occurs. The standby instance doesn't appear in the Google cloud console when failover occurs, connections to the primary instances are automatically transferred to the standby instance.

Clone --> When you clone a cloud SQL instance, you create a new instance that is a copy of the source instance but is completely independent. After cloning is complete, changes to the source instance are not reflected in the clone, and changes in the clone are not reflected in the source instance.

Replication --> Replication is the ability to create copies of a cloud SQL instance or an on-premises database, and offload work to the copies. The main reason for using replication is to scale the use of data in a database without degrading performance on the primary instance.

Database Administration

Cloud SQL lets you create and delete databases and databases users, but it is not a database administration tool. There are many database administration tools you can choose from depending on your database engine, including the following --

- a. phpMyAdmin
- b. Toad Edge
- c. MySQL workbench
- d. SSMS(SQL server management studio)

Cloud SQL pricing

- > How much storage you provision
- > How many vCPU you select
- > How much RAM you select
- > Where to choose to host your data
- > How much network traffic leaves
- > How many IP addresses you use

Cloud SQL updates

- > configuration Updates -> Google
- > System updates -> User/Administrator

Cloud Spanner

Thursday, April 24, 2025 2:12 PM

A fully managed relational database service with unlimited scale strong consistency, and up to 99.9999% Availability .

Mission critical

In AWS, AWS Aurora is Available

Cloud Spanner launched in 2017.

Provides transparent, synchronous replication across region and multi region configurations.

Optimizes performance by automatically sharding the data based on request load and size of data, so you can spend less time thinking about scaling your database and more time scaling your business.

You can run instances on a regional scope or multiple where your database is able to survive regional failure.

Used for gaming, logistics, e-commerce, insurance etc.

Cloud spanner data at least 2 TB

Billing model is hourly

Spanner Distribution

	Cloud spanner Traditional relational Traditional Non-relational		
Schema	Yes	Yes	No
SQL	Yes	Yes	No
Consistency	Strong	Strong	Eventual
Availability	High	failover	High
Scalability	Horizontal	Vertical	Horizontal
Replication	Automatic	Configurable	Configurable

ACID compliancy -Atomicity consistence isolation Durability

Pricing -->

Number of nodes in your instance

Amount of storage that your tables and secondary indexes uses

Amount of network bandwidth used

No additional charge for replication

Limitations of cloud spanner:

Lift and shift

Lots of In-database business logics (triggers and stored procedure)

When you need less latency

Cloud Big Table

Thursday, April 24, 2025 6:11 PM

Cloud Bigtable is a fully managed wide-column No SQL Databases that offer low latency and replication for high availability.

Single digit millisecond latency limit less scale and 99.999% availability for large analytical and operational workloads.

Easily migrate from Apache HBase and Cassandra to Bigtable with no downtime.

Bigtable is a sparsely populated table that can scale to billions of rows & thousands of columns.

Bigtable supports storage of terabytes or even petabytes of data.

Bigtable is not a relational database. It does not support SQL queries joins or multi-row transactions.

Data replication can be performed by simply adding a second cluster to the instance and replication starts automatically.

Bigtable integrates easily with big data tools like Hadoop, dataflow, dataproc and support HBase API's.

Bigtable stores data in Colossus Google's File system,

Scaling is horizontal scaling

Instance -

A Bigtable instance is a container for data.

Instance has one or more clusters located in a different zone and different regions.

Each cluster has at least 1 node

A table belongs to an instance and not to the cluster or node

An instance also consists of the following properties -

- a. Storage type - SSD or HDD
- b. Configuring cluster ID, Selecting region & zone.
- c. Node Scaling mode - Manual or Auto
- d. Encryption--> GMEK, CMEK
- e. Add replication cluster.

Cluster -

Cluster handles the requests sent to a single Bigtable instance.

Each cluster belongs to a single Bigtable instance and an instance can have up to 4 clusters.

Each cluster is located in a single zone.

Bigtable instances with only 1 cluster do not use replication.

Nodes -

Each Cluster in an instance has 1 or more nodes, which are the compute resources that Bigtable uses to manage the data.

Each node in the cluster handles a subset of the request to the cluster.

All client request go through a front end sever before they are sent to a Bigtable node.

Bigtable separate the compute from the storage data is never stored in nodes themselves, each node has pointer to a set of tablets they are stored on colossus.

Recovery from the failure of a Bigtable node is very fast as only the metadata needs to be migrated to the replacement node.

When a Bigtable node fails, no data is last A Bigtable cluster can be scaled out by adding nodes which would increase

The no of simultaneous req that the cluster can handle.

Bigtable nodes are sometime referred as tablet servers.

Bigtable has a limit of 1000 tables per instance

Cloud Memorystore

Thursday, April 24, 2025 7:38 PM

Memory is a fully-Managed database service that provides a managed version of two popular open source caching solution

->Redis & mem cached

Memorystore for Redis provides a fully-managed service that is powered by the Redis in-memory data store to build application caches that provide sub-millisecond data access.

Memorystore for Redis offers Several advantages over self-managed.

1. Deploy what fits your needs.
2. Easily scale to get blazing speed
3. Highly available & more secure.
4. Focus on your application.
5. Redis Protocol compatible.

It is good for caching gaming & streaming processing.

Features of memory store for Redis -

1. Fully managed.
2. Simple deployment.
3. High Availability
4. Enterprise Grade security.
5. Scale as needed.
6. Monitoring
7. On-Demand billing
8. Redis versions 6.x, 5.0, 4.0 & 3.2

You can use any standard Redis Client on the following environments to connect to your memory store for Redis Instance

1. App Engine Flexible/Standard Environment
2. Compute Engine VM instance.
3. Cloud Function
4. Cloud Run
5. GKE Clusters

Spec	Basic Tier	Standard Tier
Max Redis Node Size	300 GB	300 GB
Max Network Bandwidth	16Gbps	16Gbps
Scale Node size	yes	Yes
Cross-Zone Replication	No	Yes
Automatic Failover	No	Yes
Read Replica	No	Yes

Read Replica behavior-

Read replica are not enabled on standard tier instances by default.
Once read replica is enabled, you can't disabled it.
Standard tier instances can have 1 to 5 read replicas.

Limitations-

Read replicas are only supported for instance sizes with nodes ≥ 5 GB

Read replicas can only be enabled on instances that use Redis version 5 or higher.

Each instance has a primary endpoint and read endpoint. The primary endpoint always direct traffic to the primary node, while the read endpoint automatically load-balancer read queries across available replicas.

Failover -

When a primary fails the memorystore health monitoring service intakes the failover and the new primary is made available for both read and writes. The failover usually complete under 30 seconds.

When a failover happens the primary endpoint automatically redirect traffic to the new primary. However, all client connections to the primary endpoint one disconnected during a failover application with connect retry logic will automatically reconnect once the new primary is online.

Zone failure -

If the zone where the primary is located fails the primary automatically failover to a replica in another zone if the instance only has one replica then the read endpoint is unavailable for the duration of the zone outage if the instance has more than one replica, replicas outage of the affected zone are available for reads.

If the zone where one or more of the replicas is located failure, If there is a two zone failure and there are two or more replicas the replicas with the least lag in the remaining zone is made primary.

Maintains on Memory store for Redis -

Most of these updates are performed while your Redis instance is up and running. However, certain system updates required a brief services interruption to be performed. These updates are called maintenance. Maintenance typically falls into the following categories.

1. Memory store feature - To Launch some feature, memory store requires a maintenance update.
2. Operating system patches - We continuously monitor for newly identified security Vulnerabilities in the OS upon discovery, we patch the OS to protect you from new risks.
3. Database version upgrade - Maintenance can include an OSS Redis patch minor version update which does not include breaking or incompatible changes.

The duration of the Maintenance window is 1 hour.

If maintenance window is not specified for your instance, maintenance. Can begin at any time without advanced notification.

Version Upgrade behavior -

Data is preserved during a version upgrade operation for both basic and standard tier instances during an upgrade operation the instance undergoes a short connection reset of a couple minutes or less. The Ip address and connection string remain the same.

Import & Export data -

Import & export user cloud storage buckets to store RDB files.

An instance can only import backup from older Redis version.

Import feature behaviors -

For optimal performance you should import from cloud storage buckets located in the same region as your Redis instance

Your Redis instances is unavailable during an import operation your instance become available after the import operation finishes.

If the import is successful the existing data in the instance is overwritten by the data in the RDB file.

If the import fails, the instance is brought online but the data may be fully flushed. You can retry the import using the same RDB file or you may use another RDB file to restore data.

Resource intensive workloads -

For standard Redis Instances the following operation use extra memory for the duration of the operation.

1. Version Upgrade
2. Scaling Up/Down
3. Manual failover
4. Import / Export

All these operations and scenarios break the network connections and requires connections retry.